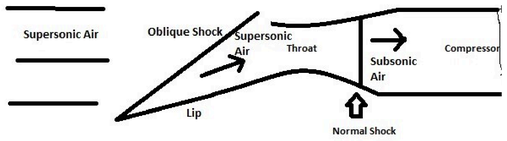
Shock Management in Supersonic Engine Inlets: Variable-Geometry vs. Fixed Designs

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Motivation



Supersonic aircraft inlets must decelerate incoming airflow to stable subsonic compressor-face conditions. Supersonic compressor inflow can cause strong shocks, pressure losses, distortion, surge, and possible engine stall

Figure 1. Figure 1: Mixed-compression supersonic inlet with oblique and normal shocks used to decelerate airflow before the compressor.



Figure 2: SR-71A reconnaissance aircraft showing inlet and fuselage configuration during flight.

At high Mach numbers, inlet design becomes increasingly difficult because:

- Shock waves cause total pressure losses
- Boundary-layer interaction may cause separation
- Flow distortion reduces engine stability
- Off-design conditions may cause the inlet unstart

Inlet performance is determined by pressure recovery, defined as **the ratio of the compressor-face total pressure to the freestream total pressure**.

Higher pressure recovery → higher thrust and efficiency
A 1% pressure recovery loss can reduce thrust by approximately 1–1.5%

Research Objective

The objective of this work is to compare two major supersonic inlet approaches used to produce acceptable compressor inflow conditions:

Variable-geometry supersonic inlets
*** Examples: Concorde ramps, SR-71 translating spike**

Fixed-geometry diverterless supersonic inlets (DSI)
*** Modern stealth-oriented bump inlets**

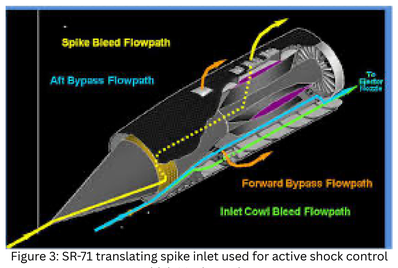


Figure 3: SR-71 translating spike inlet used for active shock control at high Mach numbers



Figure 4. Fixed-geometry diverterless supersonic inlet (DSI) using a bump surface for passive compression and boundary-layer diversion

The comparison focuses on shock management, pressure recovery, boundary-layer control, flow distortion, weight and complexity, and off-design operability.

Variable Geometry	DSI
Active shock control	Passive shock control
Higher complexity	Lower complexity
Better off-design performance	Better stealth/weight

Supersonic Inlet Fundamentals

Supersonic inlets **decelerate** and **compress** incoming airflow using oblique shocks, normal shocks, and diffuser compression before the flow reaches the compressor face

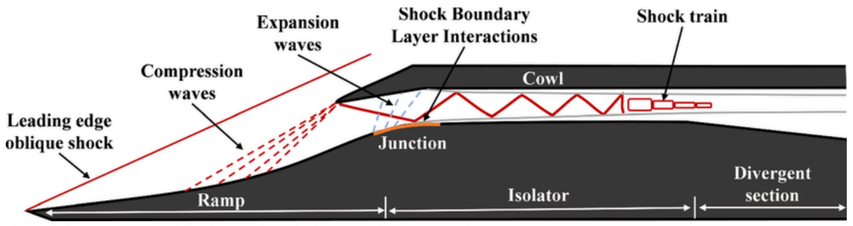


Figure 5: Shock structure and flow features in a mixed-compression supersonic inlet.

- **Oblique shocks** provide efficient compression
- **Normal shocks** produce large total pressure losses
- **Boundary-layer interaction** may cause separation
- **High compressor-face distortion** can reduce engine stability

Shock Type	Pressure Loss
Oblique Shock	Lower
Normal Shock	Higher

Variable-Geometry Supersonic Inlets

Variable-geometry inlets actively control shock position and mass flow using **movable** ramps, spikes, or throat geometries

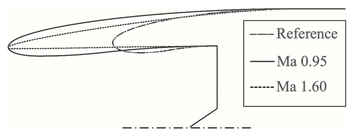


Figure 6: Optimized inlet geometries for subsonic and supersonic flight conditions

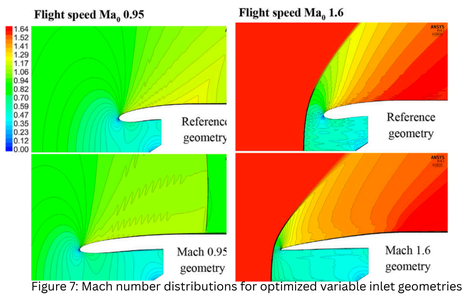


Figure 7: Mach number distributions for optimized variable inlet geometries

Variable inlet geometries adapt shape based on flight Mach number to reduce drag and improve aerodynamic performance. Figure 6 shows optimized inlet geometries for subsonic and supersonic operating conditions, while Figure 7 illustrates the resulting Mach number distributions and shock structures.

Advantages
- Improved shock control
- Higher pressure recovery
- Better off-design operability
- Enhanced compressor-face flow stability

Disadvantages
- Increased mechanical complexity
- Higher structural weight
- Greater maintenance requirements
- More difficult stealth integration

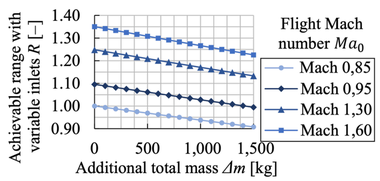


Figure 8: Range benefit of variable inlet geometries as a function of added inlet system mass

Kazula et al. found that optimized variable inlet geometries can significantly reduce aerodynamic drag, although the performance benefit decreases as inlet system mass increases (Figure 8).

Diverterless Supersonic Inlets (DSI)

Diverterless supersonic inlets (DSI) use fixed bump geometries to compress incoming supersonic airflow and divert low-energy boundary-layer flow away from the inlet entrance without movable ramps or mechanical diverters.

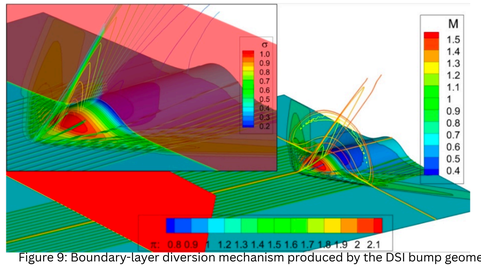


Figure 9 shows the overall geometry of a ventral DSI inlet with an integrated compression bump. Figure 10 illustrates the boundary-layer diversion mechanism produced by the bump geometry, while Figure 11 shows streamlines, pressure distribution, and Mach number contours near the inlet lip.

Figure 9: Boundary-layer diversion mechanism produced by the DSI bump geometry.

Advantages
- Reduced mechanical complexity
- Lower structural weight
- Improved stealth integration
- No moving ramps or spikes
- Simplified manufacturing and maintenance

Disadvantages
- Less adaptable across the Mach range
- More sensitive to off-design conditions
- Some low-energy flow may still enter the inlet
- Internal duct optimization remains critical
- Reduced controllability compared to variable geometry

The bump-generated oblique shock compresses the incoming flow while simultaneously producing a lateral pressure gradient that redirects low-energy boundary-layer air away from the inlet entrance. At Mach 1.6, the inlet flow is progressively compressed and decelerated before entering the duct.

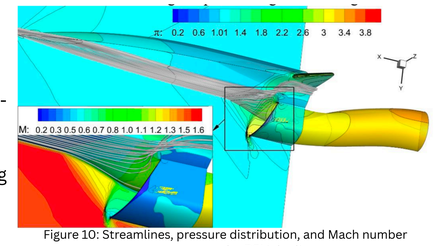


Figure 10: Streamlines, pressure distribution, and Mach number contours near the DSI inlet lip.

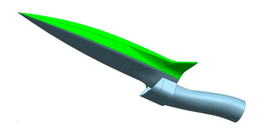


Figure 11: Geometry of a ventral diverterless supersonic inlet with an integrated compression bump.

DSI systems reduce inlet complexity by integrating shock compression and boundary-layer diversion directly into the aircraft geometry.

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